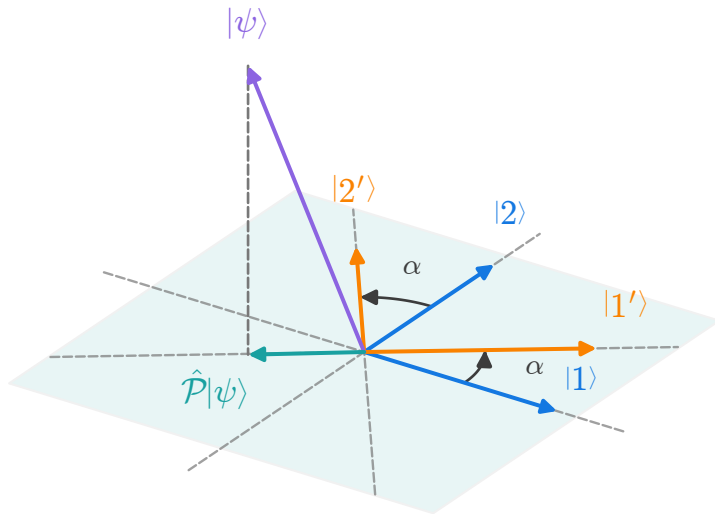




Same subspace,
different bases

$$|1'\rangle = \cos \alpha |1\rangle + \sin \alpha |2\rangle$$

$$|2'\rangle = -\sin \alpha |1\rangle + \cos \alpha |2\rangle$$

$$\alpha = \pi/5$$

$$\hat{P}' = \hat{P}$$